

MATH 102 - Applied Linear Algebra

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Fall 2025

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1 Invertible Matrix Theorem

1.1 Definition

Let A be a square $n \times n$ matrix. Then the following statements are equivalent. That is, for a given A , the statements are either all true or all false.

1. A is invertible.
2. A is row equivalent to I .

3. A has n pivot positions.
4. The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
5. The columns of A are linearly independent.
6. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
7. The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$.
8. The columns of A span $\mathbb{R}^n$.
9. The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
10. There is an $n \times n$ matrix C such that $CA = I$.
11. There is an $n \times n$ matrix D such that $AD = I$.
12. A^T is invertible.
13. The columns of A form a basis of $\mathbb{R}^n$.
14. $\text{Col } A = \mathbb{R}^n$.
15. $\dim \text{Col } A = n$.
16. $\text{rank } A = n$
17. $\text{Nul } A = \{0\}$
18. $\dim \text{Nul } A = 0$
19. The number 0 is *not* an eigenvalue of A .
20. $\det A \neq 0$.
21. $(\text{Col } A)^\perp = \{0\}$
22. $(\text{Nul } A)^\perp = \mathbb{R}^n$
23. $\text{Row } A = \mathbb{R}^n$

1.2 Thm 9.1.16

Let $A \in M_n$ and let $\lambda \in \mathbb{C}$. Then the following statements are equivalent:

- (a) λ is an eigenvalue of A .
- (b) $A\mathbf{x} = \lambda\mathbf{x}$ for some nonzero $\mathbf{x} \in \mathbb{C}^n$.
- (c) $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution, that is, $\text{nullity}(A - \lambda I) > 0$.
- (d) $\text{rank}(A - \lambda I) < n$.
- (e) $A - \lambda I$ is not invertible.
- (f) $A^\top - \lambda I$ is not invertible.
- (g) λ is an eigenvalue of $A^\top$.

Proof

- (a) $\Leftrightarrow$ (b)

By definition 9.1.1, if $A\mathbf{x} = \lambda\mathbf{x}$ and $\mathbf{x} \neq \mathbf{0}$ then $(\lambda, \mathbf{x})$ is an eigenpair of A , meaning λ is an eigenvalue of A .

- (b) $\Leftrightarrow$ (c)

These are restatements of one another.

- (c) $\Leftrightarrow$ (d)

By rank nullity theorem, if $\text{nullity}(A - \lambda I) > 0$, then $\text{rank}(A - \lambda I) < n$.

- (d) $\Leftrightarrow$ (e)

By property 1 and 16 of IMT, if $\text{rank}(A - \lambda I) < n$ then $(A - \lambda I)$ is not invertible.

- (e) $\Leftrightarrow$ (f)

Suppose we have $A - \lambda I = \begin{bmatrix} a - \lambda I & b \\ c & d - \lambda I \end{bmatrix}$, and suppose it is not invertible, its determinant is $(a - \lambda I)(d - \lambda I) - bc = 0$. Consider the case of $A^\top - \lambda I$, we have $\begin{bmatrix} a - \lambda I & c \\ b & d - \lambda I \end{bmatrix}$, whose determinant is $(a - \lambda I)(d - \lambda I) - cb$. And by commutativity of multiplication, $bc = cb$, so the determinant is also 0, meaning $A^\top - \lambda I$ is also not invertible.

(f) $\Leftrightarrow$ (g)

Since (a) is equivalent to (e), the same is true for $A^\top$.

2 Annihilator

2.1 Some Roots of an Annihilator are Eigenvalues

While all the eigenvalues of a matrix are roots of each of its annihilators, the roots of an annihilator are not necessarily eigenvalues.

This raises a question:

Must some of the roots of an annihilator of a matrix be eigenvalues of the matrix? The answer is yes.

Let $p(z) = a_0 + a_1z + a_2z^2 + \dots + a_kz^k$ be an annihilator of a matrix $A \in \mathbb{M}_n$.

Factor p as follows: $p(z) = a_k(z - \lambda_1)(z - \lambda_2) \dots (z - \lambda_k)$, $a_k \neq 0$

Since p annihilates A , we have

$$0 = p(A) = a_k(A - \lambda_1 I)(A - \lambda_2 I) \dots (A - \lambda_k I)$$

Since the zero matrix is not invertible, one of the factors $A - \lambda_i I$ is not invertible. This ensures that λ_i is an eigenvalue of A .

2.2 Each Matrix has an Eigenvector

2.2.1 Proof 1

λ is an eigenvalue of $A \in \mathbb{M}_n$ iff $\det(A - \lambda I) = 0$.

$p(\lambda) = \det(A - \lambda I)$ is a polynomial of degree n (the characteristic polynomial of A).

Therefore, λ is an eigenvalue of A iff λ is a root of p (i.e. p)

By the Fundamental Theorem of Algebra (FTA), each polynomial over the complex has a root.

Hence, each matrix $A \in \mathbb{M}_n$ has an eigenvalue.

2.2.2 Proof 2

We've shown that each matrix $A \in \mathbb{M}_n$ has an annihilator p .

By the FTA, p must have a root.

In 2.1, we showed that at least one root of p is an eigenvalue of A . Hence, each matrix $A \in \mathbb{M}_n$ has an eigenvalue.

3 Eigenvalues

3.1 Lemma 9.3.2

Let (λ, x) be an eigenpair of $A \in \mathbb{M}_n$ and let p be a polynomial. Then $p(A)x = p(\lambda)x$, that is, $(p(\lambda), x)$ is an eigenpair of $p(A)$.

Proof:

Recall $A^k \mathbf{x} = \lambda^k \mathbf{x}$.

$$\begin{aligned} p(A)\mathbf{x} &= (c_k A^k + c_{k-1} A^{k-1} + \dots + c_1 A + c_0 I)\mathbf{x} \\ &= c_k A^k \mathbf{x} + c_{k-1} A^{k-1} \mathbf{x} + \dots + c_1 A \mathbf{x} + c_0 I \mathbf{x} \\ &= c_k \lambda^k \mathbf{x} + c_{k-1} \lambda^{k-1} \mathbf{x} + \dots + c_1 \lambda \mathbf{x} + c_0 \mathbf{x} \\ &= (c_k \lambda^k + c_{k-1} \lambda^{k-1} + \dots + c_1 \lambda + c_0) \mathbf{x} \\ &= p(\lambda) \mathbf{x} \end{aligned}$$

3.2 2-Commute

Let $A_1, A_2 \in \mathbb{M}_n$. Let λ be an eigenvalue of A_1 . Suppose $A_1 A_2 = A_2 A_1$, then A_1, A_2 share an eigenvector in $\mathcal{E}_\lambda(A_1)$

Computation:

0. Choose A_1 or A_2 . Say, A_1
1. Compute an eigenvalue of A_1 . Say, λ , and form its eigenspace $\mathcal{E}_\lambda(A_1)$
2. Select a basis $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_k$ for $\mathcal{E}_\lambda(A_1)$ and form a matrix $X = [\vec{x}_1 \ \vec{x}_2 \ \dots \ \vec{x}_k]$
3. Solve $A_2 X = X C$ for C
4. Find an eigenpair of $C : (\mu, u)$
5. Compute $v = Xu$

Proof:

$$\begin{aligned} Ax_1 &= \lambda x_1, \quad Ax_2 = \lambda x_2, \quad \dots, \quad Ax_k = \lambda x_k \\ A\vec{x}_i &= \lambda \vec{x}_i \quad \text{for } i = 1, 2, \dots, k \end{aligned}$$

$$\begin{aligned} A_1 x_i &= \lambda x_i \\ A_2(A_1 x_i) &= A_2(\lambda x_i) \\ A_1(A_2 x_i) &= \lambda(A_2 x_i) \end{aligned} \quad (\text{commutativity})$$

$$\begin{aligned} \text{Case 1: } A_2 x_i &= 0 \\ \Rightarrow A_2 x_i &\in \mathcal{E}_\lambda(A_1) \end{aligned}$$

$$\begin{aligned} \text{Case 2: } A_2 x_i &\neq 0 \\ \Rightarrow A_2 x_i &\text{ is an eigenvector of } A_1 \text{ and } A_2 x_i \in \mathcal{E}_\lambda(A_1) \end{aligned}$$

$$\begin{aligned} A_2 x_1 &= X c_1, \quad A_2 x_2 = X c_2, \quad \dots, \quad A_2 x_k = X c_k \\ A_2[x_1 \ x_2 \ \dots \ x_k] &= [A_2 x_1 \ A_2 x_2 \ \dots \ A_2 x_k] = [X c_1 \ X c_2 \ \dots \ X c_k] = X[c_1 \ c_2 \ \dots \ c_k] \\ A_2 X &= X C \end{aligned}$$

$$\text{Let } (\mu, u) \text{ be an eigenpair of } C \quad C\vec{u} = \mu\vec{u} \quad u \neq 0$$

$$\text{Say } v = Xu \neq 0$$

$$Cu = \mu u$$

$$XCu = X\mu u$$

$$A_2 Xu = \mu Xu$$

$$A_2 v = \mu v$$

$$\therefore v = Xu \in \mathcal{E}_\lambda(A_1) \text{ is an eigenvector of } A_2$$

3.3 Gershgorin Disk Theorem

If $n \geq 2$ and $A = [a_{ij}] \in \mathbb{M}_n$ then

$$\operatorname{spec} A \subseteq \mathcal{G}(A) = \bigcup_{k=1}^n \mathcal{G}_k(A).$$

Proof:

Let $(\lambda, \mathbf{x})$ be an eigenpair of $A = [a_{ij}] \in \mathbb{M}_n$ and let $\mathbf{x} = [x_i]$. Since $\lambda \mathbf{x} = A\mathbf{x}$, we have

$$\begin{aligned} \lambda x_i &= \sum_{j=1}^n a_{ij} x_j = a_{ii} x_i + \sum_{j \neq i} a_{ij} x_j \\ (\lambda - a_{ii}) x_i &= \sum_{j \neq i} a_{ij} x_j \end{aligned}$$

Let $k \in \{1, 2, \dots, n\}$ be any index such that $|x_k| = \max_{1 \leq i \leq n} |x_i|$. Since $\mathbf{x} \neq \mathbf{0}$, it follows that